

LYNGOR FYR, Norway

WMO: 014670

Lat: 58.636N

Lon: 9.148E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.4	-7.5	-14.6	1.1	-5.3	-12.6	1.3	-4.2	15.3	1.1	14.1	2.1	4.3	10	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	22.9	17.7	21.8	17.0	20.7	16.3	19.0	21.7	18.1	20.7	17.2	19.7	4.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.8	20.8	16.9	12.0	19.8	16.0	11.4	18.9	53.9	21.9	51.0	20.9	48.3	19.8	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.8	10.4	-10.5	25.1	2.3	1.9	-12.2	26.4	-13.5	27.5	-14.8	28.6	-16.5	29.9		
(5)			-11.0	19.7	2.2	1.0	-12.6	20.4	-13.9	21.0	-15.1	21.6	-16.7	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	0.8	0.5	2.6	6.2	10.8	14.6	17.2	16.9	13.6	9.0	5.1	1.9		
	DBStd	6.76	3.91	4.04	3.35	2.70	2.98	2.32	2.13	2.04	2.11	2.76	3.45	4.36		
	HDD10.0	1375	285	266	229	118	23	0	0	0	2	50	150	252		
	HDD18.3	3683	543	499	487	364	233	114	48	56	142	288	398	511		
	CDD10.0	762	0	0	1	4	49	138	224	213	110	21	2	0		
	CDD18.3	28	0	0	0	0	1	3	14	11	0	0	0	0		
	CDH23.3	27	0	0	0	0	1	3	18	4	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.8	5.5	5.2	4.9	4.3	4.4	4.4	4.2	4.4	4.9	5.2	5.4	5.0		
(15) Precipitation	PrecAvg	1164	112	75	49	64	74	78	88	127	117	136	136	100		
	PrecMax	1594	288	227	127	179	174	174	194	219	295	341	465	182		
	PrecMin	789	13	8	0	6	7	7	19	23	46	21	36	36		
	PrecStd	211	68	48	37	41	40	40	45	59	68	74	91	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	10.0	13.3	15.9	21.6	23.3	25.4	23.9	20.1	15.7	13.0	10.7		
		MCWB	6.7	6.0	7.7	10.3	14.7	15.5	18.5	18.1	14.9	12.0	10.9	9.2		
	2%	DB	7.7	8.1	9.9	13.3	18.9	21.2	23.3	22.6	18.7	14.6	11.8	9.4		
		MCWB	5.6	4.9	5.6	8.6	13.3	14.7	17.9	18.2	14.4	12.0	10.1	7.9		
	5%	DB	6.7	6.7	8.1	11.6	16.8	19.6	22.0	21.6	17.7	13.7	10.7	8.5		
		MCWB	5.2	4.6	4.3	7.4	12.0	14.1	17.2	17.5	14.3	11.6	9.1	7.1		
	10%	DB	5.8	5.7	6.7	10.3	15.2	18.2	21.0	20.5	16.9	13.0	9.7	7.5		
		MCWB	4.4	4.2	3.9	6.8	11.0	13.5	16.7	16.6	14.1	11.1	8.2	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	6.8	8.3	11.4	15.7	17.6	20.0	20.3	17.3	13.9	11.6	9.6		
		MCDB	8.3	8.3	12.7	14.7	19.9	20.8	23.1	22.5	18.6	14.8	12.4	10.1		
	2%	WB	6.3	5.9	6.2	9.2	14.3	16.1	19.0	19.3	16.2	12.9	10.4	8.4		
		MCDB	7.3	7.0	8.7	12.0	17.6	19.3	22.0	21.7	17.5	13.8	11.4	9.2		
	5%	WB	5.5	5.1	5.3	8.3	13.0	15.3	18.1	18.3	15.5	12.2	9.4	7.3		
		MCDB	6.4	6.4	7.0	10.7	15.8	18.2	21.0	20.8	16.9	13.2	10.5	8.2		
	10%	WB	4.7	4.2	4.6	7.4	11.8	14.5	17.3	17.4	14.7	11.5	8.4	6.2		
		MCDB	5.6	5.4	6.1	9.4	14.3	17.2	20.1	19.8	16.3	12.7	9.6	7.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	4.0	4.6	4.7	4.8	4.8	4.8	4.7	4.5	4.1	3.7	3.7		
		MCDBR	4.6	5.3	7.0	6.8	6.8	6.7	6.4	5.3	5.5	4.6	4.6	4.3		
	5% WB	MCWBR	4.2	3.7	4.3	4.0	3.9	4.2	3.9	3.7	3.9	3.7	4.4	4.2		
		MCWBR	4.0	4.3	6.0	5.4	5.7	5.1	5.0	4.6	4.3	3.6	4.3	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.321	0.340	0.379	0.372	0.357	0.378	0.377	0.351	0.336	0.304	0.266		
		taud	2.242	2.320	2.366	2.325	2.417	2.481	2.441	2.439	2.503	2.447	2.302	2.181		
	Edn at Noon		537	703	803	826	861	880	850	822	791	684	514	430		
		Edh at Noon	46	69	87	107	104	99	101	95	76	61	43	34		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.91	2.35	3.90	5.21	5.94	5.39	4.16	2.69	1.22	0.45	0.23		
	RadStd		0.07	0.19	0.21	0.55	0.59	0.43	0.55	0.26	0.33	0.20	0.12	0.03		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.60	N/A	N/A	-1.05	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page